Uncovering Employee Work–Wellbeing Profiles via Unsupervised Clustering

A PROJECT REPORT

Submitted by

Batoul Hamieh

Samira Jawish

in partial fulfillment of the requirements for the Data Mining course

Lebanese University

Supervisor:

Dr. Abbas Rammal

January 2026

TABLE OF CONTENTS

1.0	**INTRODUCTION**	**4**
2.0	**DATASET DESCRIPTION**	**4**
3.0	**METHODOLOGY**	**5**
4.0	**EXPLORATORY DATA ANALYSIS**	**7**
5.0	**DATA PREPROCESSING**	**13**
6.0	**FEATURE ENGINEERING**	**15**
7.0	**K-MEANS CLUSTERING**	**17**
8.0	**CLUSTERING EVALUATION**	**19**
9.0	**CLUSTER LABELING**	**23**
10.0	**CONCLUSION**	**27**

LIST OF FIGURES

Figure 1	**Distribution of Work Arrangement**	**11**
Figure 2	**KMeans Inertia for Different K Values**	**20**
Figure 3	**Silhouette for Different K Values**	**21**
Figure 4	**3D Scatter Plot of Employee Data by Cluster**	**22**
Figure 5	**Work Modality Intensity by Cluster**	**23**
Figure 6	**Psychophysiological Health Strain Score by Cluster**	**24**
Figure 7	**Perceived Well Being Social Index by Cluster**	**24**

1. INTRODUCTION

This study is based on the Post-Pandemic Work Health Impact 2025 dataset, which was originally designed to examine the effects of remote work on employee well-being, social isolation, and burnout. The dataset aimed to provide insights into how adopting remote work as a regular part of professional life influences work–life balance, taking into account demographic and occupational differences.

Through analysis, however, it became evident that the factors affecting employee well-being extend beyond the type of work arrangement alone. Other external and workplace-related factors, including workload, social isolation, industry, job role, and mental and physical health, also play a significant role in shaping overall employee health and social outcomes. Consequently, the study's focus shifted from solely assessing work arrangement to understanding the broader combination of factors that contribute to employee well-being and social experiences. The research ultimately investigates the extent to which different aspects of work, beyond just modality, influence employees' mental, physical, and social health in the post-pandemic environment by segmenting the states at which employees are performing best, worst and not as effective through K-Means clustering.

2. DATASET DESCRIPTION

The Remote Work Health Impact Survey 2025 dataset is a cross-sectional survey collected in **June 2025** to provide a comprehensive perspective on how post-pandemic work arrangements influence employee health, well-being, and work outcomes. The dataset was gathered globally and includes responses from a diverse workforce spanning multiple regions, industries, age groups, and job roles. It is designed to support research, data analysis, and policy discussions around the evolving nature of work and its implications for employee health and productivity.

Key Characteristics

- **Collection Period:** June 2025
- **Scope:** Global respondents across continents and occupational sectors
- **Purpose:** To explore the relationships between work arrangements (remote, hybrid, onsite) and employee mental, physical, social, and productivity outcomes
- **Structure:** Survey responses include a mix of demographic, workplace, health, and well-being variables, enabling multidimensional analysis

Primary Variable Categories

- **Demographic Attributes**: Age, Gender, Region, Industry, Job role
- **Workplace Factors:** Work arrangement (onsite, hybrid, remote), Hours worked per week, Work–life balance scores, Burnout level
- **Health and Well-Being Indicators**: Mental health status (anxiety, burnout, PTSD, depression), Physical health complaints (shoulder pain, neck pain, back pain, eye strain, wrist pain), Social isolation scores
- **Additional Outcomes:** Salary range and other measures related to perceived well-being, productivity trends, and lifestyle factors

Analytical Potential

- Enables assessment **of** prevalence and distribution of mental and physical health conditions across different work arrangements
- Supports exploration of relationships between work modality and well-being outcomes, such as burnout and social isolation
- Facilitates segmentation of employee groups for pattern discovery using unsupervised learning methods
- Provides a foundation for evaluating interactions between demographic, workplace, and health variables

3. METHODOLOGY

Exploratory Data Analysis (EDA)

The study began with exploratory data analysis to understand the structure, distribution, and quality of the survey data. Descriptive statistics were computed for both numerical and categorical variables to assess central tendencies, variability, and the presence of null or unique values. Particular attention was given to mental and physical health indicators, as these contained multi-select responses and null entries reflecting survey design rather than missing data. EDA confirmed that the dataset was clean, with all variables properly formatted and no anomalies detected, providing a reliable foundation for subsequent analysis.

Data Cleaning and Preprocessing

Non-behavioral and identifying features, including demographic and organizational variables, were removed to ensure clustering focused on meaningful patterns related to employee experience and well-being. Categorical variables were encoded numerically: mental health conditions, physical health issues, and work arrangements were transformed using binary one-hot encoding, while burnout levels were represented through ordinal label encoding. Numerical features such as hours worked per week, burnout level, work–life balance, and social isolation scores were standardized using scaling to prevent disproportionate influence during distance-based clustering, ensuring that each feature contributed equitably to the analysis.

Feature Engineering

In this phase, the raw survey data was transformed into meaningful measures representing employee work, health, and well-being. Work arrangements were replaced to reflect both modality and workload, capturing the interaction between work type and intensity. Mental and physical health indicators were aggregated into composite severity and strain scores to reflect cumulative well-being, while work–life balance and social isolation were combined into a perceived well-being index. These transformations reduced dimensionality, harmonized related variables, and produced interpretable features that highlight the complex interactions influencing employee experiences, providing a robust foundation for clustering analysis.

K-Means Clustering

Clustering was conducted using the K-Means algorithm to identify behavioral profiles within the employee dataset. The analysis was performed on three engineered composite features: Work Modality Intensity, Psychophysiological Health Strain Score, and Perceived Well-Being Social Index, selected to represent structural work characteristics, cumulative health strain, and psychosocial well-being respectively. K-Means was chosen due to its suitability for continuous, low-dimensional data and its interpretability in exploratory behavioral segmentation. An exploratory range of cluster solutions was evaluated, with the number of clusters K varying from 2 to 14 to allow the underlying structure of the data to emerge empirically rather than imposing an a priori segmentation.

Clustering Evaluation

To determine the optimal number of clusters, multiple internal validation metrics were employed. For each candidate value of K, inertia was computed to assess cluster compactness, while silhouette scores were calculated to evaluate the balance between within-cluster cohesion and between-cluster separation. The convergence of these criteria supports the selection of K=3 as a statistically sound solution. Visual inspection through a three-dimensional scatter plot of the standardized feature space further corroborated this choice with clear spatial separation among clusters.

Cluster Labeling

Violin plots were used to examine the distribution of each engineered feature across clusters and enable comparison of central tendencies, dispersion, and distributional shape. Based on these patterns, clusters were labeled to reflect distinct employee experience profiles: *Thriving*, *Detached*, and *Vulnerable*. This step transforms abstract numerical groupings into actionable profiles that supports downstream application in policy design, intervention targeting, and strategic workforce planning.

4. EXPLORATORY DATA ANALYSIS

In this phase of the data science life cycle, the study focuses on exploring and understanding the underlying structure of the survey data in order to assess the impact of various work arrangements, lifestyle factors, and social conditions on workers across different job roles, age groups, and industries. The objective of this stage is to identify initial patterns, distributions, and potential relationships that may inform subsequent data preprocessing and modeling steps.

To conduct the exploratory data analysis, the Python library **pandas** was utilized to efficiently load, manipulate, and inspect the dataset in a structured and dynamic format, enabling systematic examination of variables and facilitating data-driven insights.

Dataset Overview:

The data was consisted of 3,157 rows and 14 columns (features), indexed as:

Among all the features it has 10 objects and 5 numerical columns and as it is observed almost all the features have an equal number of non-null values (3,157) except for the Physical_Health_Issues with only

2,877 rows and the Mental_Health_Status with 2,358 rows.

For the assigned numerical columns, the .describe() function was used to showcase the statistical summary of these data across the count,

mean, standard deviation, minimum and maximum values also through quartiles.

Data Quality Assessment:

[7]:

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 3157 entries, 0 to 3156
Data columns (total 14 columns):
 #   Column                   Non-Null Count  Dtype
---  ------                   --------------  -----
 0   Survey_Date              3157 non-null   object
 1   Age                      3157 non-null   int64
 2   Gender                   3157 non-null   object
 3   Region                   3157 non-null   object
 4   Industry                 3157 non-null   object
 5   Job_Role                 3157 non-null   object
 6   Work_Arrangement         3157 non-null   object
 7   Hours_Per_Week           3157 non-null   int64
 8   Mental_Health_Status     2358 non-null   object
 9   Burnout_Level            3157 non-null   object
 10  Work_Life_Balance_Score  3157 non-null   int64
 11  Physical_Health_Issues   2877 non-null   object
 12  Social_Isolation_Score   3157 non-null   int64
 13  Salary_Range             3157 non-null   object
dtypes: int64(4), object(10)
memory usage: 345.4+ KB
```

Columns Detected with Null Values:

- Mental_Health_Status
- Physical_Health_Issues

The descriptive statistics provide an overview of the demographic and work-related characteristics of the 3,157 respondents and indicate a dataset that is structurally complete and well-formed. All variables exhibit full observation counts, with no missing values or anomalous ranges detected. The age distribution centers around a mean of approximately 44 years, while weekly working hours average close to 50 hours, reflecting a workforce with varied career stages and workload intensities.

Perceived work–life balance and social isolation scores are distributed around the midpoint of their respective five-point scales, with values spanning the full range without evidence of irregular or extreme distortions. Overall, the consistency of the distributions and the absence of unexpected values showcase that the data is clean and reliable, providing a stable foundation for subsequent feature engineering and clustering analysis.

[67]:

```
df.describe()
```

[67]:

	Age	Hours_Per_Week	Work_Life_Balance_Score	Social_Isolation_Score
count	3157.00	3157.00	3157.00	3157.00
mean	43.73	49.90	3.00	2.70
std	12.66	8.90	1.16	1.19
min	22.00	35.00	1.00	1.00
25%	33.00	42.00	2.00	2.00
50%	44.00	50.00	3.00	3.00
75%	55.00	57.00	4.00	4.00
max	65.00	65.00	5.00	5.00

Observations:

- Ordinal variables (1-5): "Work_Life_Balance_Score", "Social_Isolation_Score"
- numerical variables: "Age"(22–65), "Hours_Per_Week"(35–65)
- Needs scaling

```
df.describe(include='O')
```

[9]:

	Survey_Date	Gender	Region	Industry	Job_Role	Work_Arrangement	Mental_Health_Status	Bur
count	3157	3157	3157	3157	3157	3157	2358	
unique	26	4	6	9	24	3	6	
top	2025-06-07	Male	South America	Professional Services	Research Scientist	Onsite	PTSD	
freq	137	1535	575	730	151	1562	423	

The categorical variable summary indicates a well-structured dataset with consistent encoding across demographic, occupational, and work arrangement attributes. All categorical fields exhibit complete observation counts, with the exception of *Mental_Health_Status* and *Physical_Health_Issues*, where lower counts reflect non-responses rather than data irregularities. The number of unique values per variable remains within reasonable bounds, suggesting controlled category definitions without excessive fragmentation.

The most frequent categories indicate a workforce primarily composed of male respondents, predominantly located in South America, with strong representation in professional services and onsite work arrangements. Importantly, no unexpected or malformed category values are observed, and frequency distributions appear balanced enough to support meaningful subgroup analysis.

<u>Unique and Null Values:</u>

```
[11]: df["Mental_Health_Status"].unique()

[11]: array(['Stress Disorder', 'ADHD', nan, 'Burnout', 'Anxiety', 'PTSD',
       'Depression'], dtype=object)

[12]: df["Work_Arrangement"].unique()

[12]: array(['Onsite', 'Hybrid', 'Remote'], dtype=object)

[13]: df["Burnout_Level"].unique()

[13]: array(['High', 'Medium', 'Low'], dtype=object)
```

An examination of unique and null values reveals that most categorical variables are clearly and consistently defined. Mental health status is represented by a fixed set of categories, including stress disorder, ADHD, burnout, anxiety, PTSD, and depression, alongside null values. These null entries do not indicate data corruption but rather reflect respondents who did not identify with any of the listed conditions, as the available options capture only specific mental health states without a neutral alternative. A similar pattern is observed for physical health issues, where missing values indicate the absence of reported conditions rather than incomplete responses.

[10]:

```
df["Physical_Health_Issues"].unique()
```

[10]:

```
array(['Shoulder Pain; Neck Pain', 'Back Pain',
       'Shoulder Pain; Eye Strain', nan, 'Back Pain; Shoulder Pain',
       'Back Pain; Shoulder Pain; Wrist Pain', 'Neck Pain',
       'Shoulder Pain', 'Eye Strain; Wrist Pain', 'Back Pain; Eye Strain',
       'Back Pain; Shoulder Pain; Eye Strain', 'Back Pain; Wrist Pain',
       'Shoulder Pain; Eye Strain; Neck Pain; Wrist Pain', 'Eye Strain',
       'Back Pain; Shoulder Pain; Eye Strain; Neck Pain',
       'Eye Strain; Neck Pain', 'Shoulder Pain; Eye Strain; Wrist Pain',
       'Back Pain; Neck Pain', 'Wrist Pain',
       'Back Pain; Shoulder Pain; Neck Pain',
       'Back Pain; Eye Strain; Neck Pain',
       'Back Pain; Shoulder Pain; Eye Strain; Wrist Pain',
       'Shoulder Pain; Eye Strain; Neck Pain', 'Neck Pain; Wrist Pain',
       'Back Pain; Eye Strain; Neck Pain; Wrist Pain',
       'Back Pain; Eye Strain; Wrist Pain',
       'Back Pain; Neck Pain; Wrist Pain', 'Shoulder Pain; Wrist Pain',
       'Back Pain; Shoulder Pain; Eye Strain; Neck Pain; Wrist Pain',
       'Back Pain; Shoulder Pain; Neck Pain; Wrist Pain',
       'Eye Strain; Neck Pain; Wrist Pain',
       'Shoulder Pain; Neck Pain; Wrist Pain'], dtype=object)
```

Notably, physical health issues are recorded as concatenated multi-select responses, resulting in 31 unique values that represent combinations of conditions rather than distinct issues. For example, shoulder pain, neck pain, and back pain may appear together within a single entry, separated by semicolons.

While these encoding preserves survey responses, it disregards the uniqueness of each physical issue requiring further preprocessing to decompose these combinations into separate states of physical pain.

Distributions:

The distribution of work arrangements indicates that onsite employment remains the dominant mode, with approximately 1,560 respondents reporting onsite work, exceeding both hybrid workers (approximately 1,000) and remote workers (approximately 700). This distribution suggests that, despite the post-pandemic expansion of flexible work models, traditional onsite work continues to be the preferred/admitted arrangement within the sampled workforce.

[16]:

```
plt.figure(figsize=(6,4))
sns.countplot(data=df, x="Work_Arrangement")
plt.title("Distribution of Work Arrangement")
plt.xlabel("Work Arrangement")
plt.ylabel("Number of Employees")
plt.show()
```

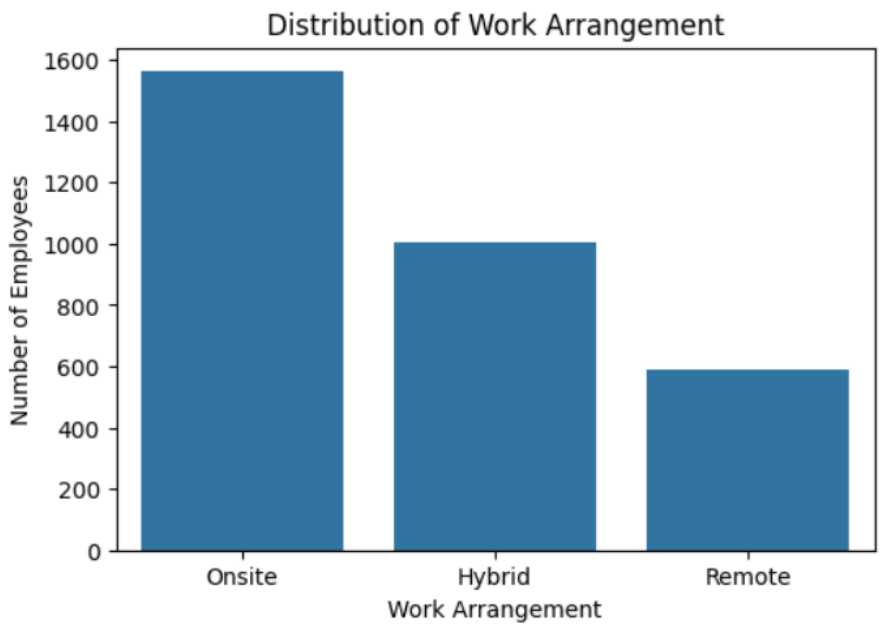

Figure 1: Distribution of Work Arrangement

Columns Excluded from Clustering Analysis:

- "Survey_Date"
- "Region"
- "Industry"
- "Job_Role"
- "Gender"
- "Age"
- "Salary_Range"

Rationale: These columns are not behavioral indicators of employee experience, productivity, or well-being. Including them in the clustering process could create clusters driven by identifiers rather than meaningful patterns.

However, these variables can still be useful for post-clustering profiling.

To conclude the exploratory insights, features that are primarily demographic, organizational, were excluded from the clustering analysis, as they do not directly represent behavioral patterns related to employee experience, productivity, or well-being. Specifically, variables such as *Survey_Date, Region, Industry, Job_Role, Gender, Age,* and *Salary_Range* were removed to prevent the formation of clusters driven by personal or contextual identifiers rather behavioral interactions. Nonetheless, these excluded attributes retain analytical value and were preserved for post-clustering interpretation.

5. DATA CLEANING

One-Hot and Label Encoding:

Following the exploratory analysis, the dataset was subjected to systematic cleaning. Columns identified as non-behavioral or personal:*Survey_Date, Region, Job_Role, Industry, Gender, Age,* and *Salary_Range*, were removed to ensure that subsequent clustering focused exclusively on variables reflecting employee experience, productivity, and well-being.

[18]:

```
cleaned_df["work_onsite"] = 0
cleaned_df["work_hybrid"] = 0
cleaned_df["work_remote"] = 0
```

[19]:

```
cleaned_df.loc[cleaned_df["Work_Arrangement"] == "Onsite","work_onsite"] = 1
cleaned_df.loc[cleaned_df["Work_Arrangement"] == "Hybrid","work_hybrid"] = 1
cleaned_df.loc[cleaned_df["Work_Arrangement"] == "Remote","work_remote"] = 1
```

[20]:

```
cleaned_df.drop("Work_Arrangement", axis=1, inplace=True)
```

[21]:

```
cleaned_df["has_Stress_Disorder"] = 0
cleaned_df["has_ADHD"] = 0
cleaned_df["has_Burnout"] = 0
cleaned_df["has_Anxiety"] = 0
cleaned_df["has_PTSD"] = 0
cleaned_df["has_Depression"] = 0
```

[22]:

```
cleaned_df.loc[cleaned_df["Mental_Health_Status"] == "Stress Disorder", "has_Stress_Disorder"] = 1
cleaned_df.loc[cleaned_df["Mental_Health_Status"] == "ADHD", "has_ADHD"] = 1
cleaned_df.loc[cleaned_df["Mental_Health_Status"] == "Burnout", "has_Burnout"] = 1
cleaned_df.loc[cleaned_df["Mental_Health_Status"] == "Anxiety", "has_Anxiety"] = 1
cleaned_df.loc[cleaned_df["Mental_Health_Status"] == "PTSD","has_PTSD"] = 1
cleaned_df.loc[cleaned_df["Mental_Health_Status"] == "Depression", "has_Depression"] = 1
```

[23]:

```
cleaned_df.drop("Mental_Health_Status", axis=1, inplace=True)
```

Categorical features were then encoded to enable numerical analysis. For mental health status and physical health issues, each distinct condition was transformed into a separate binary column, where a value of 1 indicated the presence of the condition and 0 indicated its absence. Similarly, work arrangement categories (onsite, remote, hybrid) were encoded as individual binary columns. In contrast, the burnout level, which follows an ordinal scale, was label-encoded with low, medium, and high mapped to 1, 2, and 3, respectively. These transformations ensured that all features were numerically represented and suitable for both clustering and downstream modeling. Then, the Work Arrangement, Mental_Health_Status, Physical_Health_Issues were dropped.

```
[29]: cleaned_df.head()
```

[29]:

has_Stress_Disorder	has_ADHD	has_Burnout	has_Anxiety	has_PTSD	has_Depression	has_Shoulder_Pain	has_Neck_Pain	has_Back_Pain	has_Eye_Strain	has_Wrist_Pain
1	0	0	0	0	0	1	1	0	0	0
1	0	0	0	0	0	0	0	1	0	0
0	1	0	0	0	0	1	0	0	1	0
0	1	0	0	0	0	1	0	0	1	0
0	0	0	0	0	0	0	0	0	0	0

<u>Scaling:</u>

In addition to encoding categorical variables, all numerical features were scaled to standardize their ranges and ensure comparability across variables. Specifically, the columns *Hours_Per_Week, Burnout_Level, Work_Life_Balance_Score,* and *Social_Isolation_Score* were transformed using standard scaling, which centers the values around zero with unit variance. Scaling is necessary because clustering algorithms, such as KMeans, rely on distance-based measures; without standardization, features with larger numeric ranges could disproportionately influence cluster formation, potentially biasing the results. By scaling these variables, each feature contributes equally to the distance calculations, enabling more balanced and meaningful clustering outcomes.

```
[30]: scaler = StandardScaler()
      scaled_data = scaler.fit_transform(cleaned_df[["Hours_Per_Week", "Burnout_Level", "Work_Life_Balance_Score","Social_Isolation_Score"]])
      scaled_data
```

```
[30]: array([[ 1.58437147,  1.21082339,  0.00299566, -0.59290285],
             [-1.45060178,  1.21082339,  0.86275015, -0.59290285],
             [-1.5630082 ,  1.21082339,  0.00299566, -0.59290285],
             ...,
             [ 1.02233938, -0.12761129, -1.71651333,  0.24835301],
             [ 0.23549447, -0.12761129,  0.00299566,  1.93086473],
             [-1.22578895, -0.12761129,  0.86275015, -0.59290285]],
            shape=(3157, 4))
```

6. FEATURE ENGINEERING

Work Modality Score

To replace the work arrangement information, an ordinal *Work_Modality_Score* was created using the equation:

$$Work_{Modality_{Score}} = (work_{onsite} \times 0) + \left(work_{hybrid} \times 1\right) + (work_{remote} \times 2)$$

Here, the binary columns indicating onsite, hybrid, and remote work were mapped to integers reflecting increasing degrees of remote work. This transformation reduces dimensionality from three separate columns into a single, interpretable measure, allowing clustering algorithms to capture patterns related to employee work modality without being influenced by redundant binary indicators.

Work Modality Intensity

To integrate work arrangement with workload, a *Work_Modality_Intensity* variable was constructed as the mean of the work modality score and weekly hours worked:

$$Work_{Modality_{Intensity}} = \frac{Work_{Modality_{Score}} + Hours_Per_Week}{2}$$

This feature combines the type of work (onsite, hybrid, remote) with the actual time commitment, providing a clear measure of work intensity. By capturing both the modality and workload, the index enhances clustering's ability to differentiate between employees with similar arrangements but different work demands.

Mental Health Severity Score

A *Mental_Health_Severity_Score* was derived to represent overall mental health impact using the following aggregation:

$$\begin{aligned} Mental_{Health_{Severity_{Score}}} &= \left(has_{Stress_{Disorder}}\right) \times 1 + (has_{ADHD}) \times 1 + \left(has_{Anxiety}\right) \times 2 + (has_{Burnout}) \times 3 \\ &+ \left(has_{Depression} \times 4\right) + (has_PTSD \times 4) \end{aligned}$$

Here, each mental health condition was weighted according to its perceived severity (0 = no issue, up to 4 = depression/PTSD). This creates a single ordinal measure that captures the cumulative effect of mental health conditions, allowing clustering to consider mental well-being as a continuous severity dimension rather than multiple separate indicators.

Physical Strain Score

Physical health burden was summarized with a *Physical_Strain_Score*, calculated as the sum of individual physical health conditions:

$$Physical_{Strain_{Score}} = has_{Shoulder_{Pain}} + has_{Neck_{Pain}} + has_{Back_{Pain}} + has_{Eye_{Strain}} + has_Wrist_Pain$$

This aggregation captures the total physical strain experienced by an employee, combining multiple conditions into one meaningful measure. It reduces dimensionality while preserving the information about cumulative physical health impact.

Psychophysiological Health Strain Score

To capture the combined impact of both mental and physical health on employees, a composite *Psychophysiological_Health_Strain_Score* was constructed by averaging the previously derived mental and physical strain measures:

$$Psychophysiological_{Health_{Strain_{Score}}} = \frac{Mental_{Health_{Severity_{Score}}} + Physical_Strain_Score}{2}$$

This new feature integrates psychological and physiological dimensions into a single continuous variable, providing a measure of overall health-related strain. By combining these two aspects, the score reduces dimensionality and allows clustering algorithms to account for the cumulative burden of both mental and physical conditions, while preserving interpretability. Following the creation of this composite score, the individual mental and physical strain columns were removed to avoid redundancy.

Inverted Work–Life Balance Score

To align directionality with other strain indicators, the *Work_Life_Balance_Score* was inverted:

$$Work_{Life_{Balance_{Score_{Inverted}}}} = -1 \times Work_Life_Balance_Score$$

By doing so, higher values now indicate worse balance, consistent with burnout and social isolation measures. This standardization ensures that all features representing strain or stress are oriented in the same direction, improving interpretability and aggregation.

Perceived Well-Being Social Index

Finally, a composite *Perceived_WellBeing_SocialIndex* was created to capture overall employee well-being:

$$Precieved_{WellBeing_{SocialIndex}} = \frac{Burnout_{Level} + Work_{Life_{Balance_{Score_{Inverted}}}} + Social_Isolation_Score}{3}$$

This index averages burnout, inverted work–life balance, and social isolation scores to provide a single continuous measure of perceived well-being. By integrating multiple correlated measures, the index facilitates clustering based on overall psychological and social health, while maintaining interpretability for later analysis.

7. K MEANS CLUSTERING

Following feature engineering, clustering was conducted using three composite behavioral features designed to capture the central dimensions of work arrangement experience:

- **Work_Modality_Intensity:** reflecting the intensity of on-site, hybrid, and remote work engagement.
- **Psychophysiological_Health_Strain_Score:** representing cumulative mental and physical health strain associated with work conditions.
- **Perceived_WellBeing_SocialIndex:** capturing perceived well-being through indicators of work–life balance and social isolation.

Prior to clustering, all features were standardized using z-score normalization to ensure comparable scales across dimensions, as K-Means relies on distance-based optimization and would otherwise be disproportionately influenced by variables with larger numeric ranges.

K-Means Objective Function:

K-Means clustering partitions the dataset into K clusters by minimizing the within-cluster sum of squared distances between observations and their assigned cluster centroids. Formally, the objective function is defined as:

$$\sum_{k=1}^{K} \sum_{x_i \in C_k} \|x_i - \mu_k\|^2$$

where:

- $\mathbf{C_k}$ denotes the set of observations assigned to cluster k
- $\mathbf{X_k}$ represents an individual observation
- $\boldsymbol{\mu_k}$ is the centroid of cluster k
- $\|\cdot\|^2$ denotes squared Euclidean distance.

The algorithm iteratively alternates between assigning observations to the nearest centroid and updating centroids as the mean of assigned points, until convergence.

Euclidean Distance Metric:

K-Means clustering inherently relies on Euclidean distance to quantify similarity between observations and cluster centroids. This metric is appropriate in the present context, as all input features are continuous, standardized, and conceptually orthogonal dimensions.

However, the use of Euclidean distance introduces several implicit assumptions:

- Clusters are approximately spherical in feature space
- Clusters exhibit comparable variance across dimensions
- Clusters are meaningfully separated along linear combinations of the features.

While behavioral data rarely conform perfectly to these assumptions, K-Means remains a methodologically sound choice for exploratory segmentation that provides interpretable clusters even in the presence of modest deviations from idealized geometric conditions.

Selection of Candidate K Values:

Rather than imposing a predetermined number of clusters, a data-driven exploratory range of **K=2** to **K=14** was considered. This range balances two competing considerations: smaller values of K risk oversimplifying the inherent heterogeneity of behavioral patterns, while larger values may produce fragmented clusters that are challenging to interpret meaningfully. Evaluating this range empirically allows the cluster structure to emerge from the data, with the final selection of the optimal number of clusters determined through formal validation.

Practical Implementation Details:

K-Means clustering was executed using an iterative optimization procedure with specified settings:

- **Centroid initialization:** employed the algorithm's default strategy.
- **Randomness:** controlled by fixing the random seed (random_state = 42) for guaranteed consistent results across repeated runs.
- **Convergence parameters:** allowed up to 1,000 iterations per run to reach stable solutions even under less favorable initializations.

```
max_k=14
inertia=[]
silhouette_scores=[]
k_values=range(2,max_k + 1)

for k in k_values:
    kmeans= KMeans(n_clusters=k, random_state=42,max_iter=1000)
    cluster_labels=kmeans.fit_predict(final_df)
    sil_score= silhouette_score(final_df,cluster_labels)
    silhouette_scores.append(sil_score)
    inertia.append(kmeans.inertia_)
```

8. CLUSTERING EVALUATION

In unsupervised learning, cluster labels are not externally defined, and no ground truth exists against which model performance can be directly measured. Consequently, determining the appropriate number of clusters requires internal validation techniques that assess how well the resulting partition balances inter-cluster separation.

To address this, the clustering solutions were evaluated across a range of candidate values using complementary validation metrics:

- **Inertia (Within-Cluster Sum of Squares):** used in the Elbow Method to assess cluster compactness.
- **Silhouette Score:** used to evaluate the degree of separation between clusters relative to within-cluster cohesion.

Elbow Method:

Inertia measures the total squared Euclidean distance between observations and their assigned cluster centroids. As K increases, inertia monotonically decreases; therefore, its absolute minimum is not informative. Instead, the Elbow Method seeks a point beyond which additional clusters yield diminishing returns in compactness.

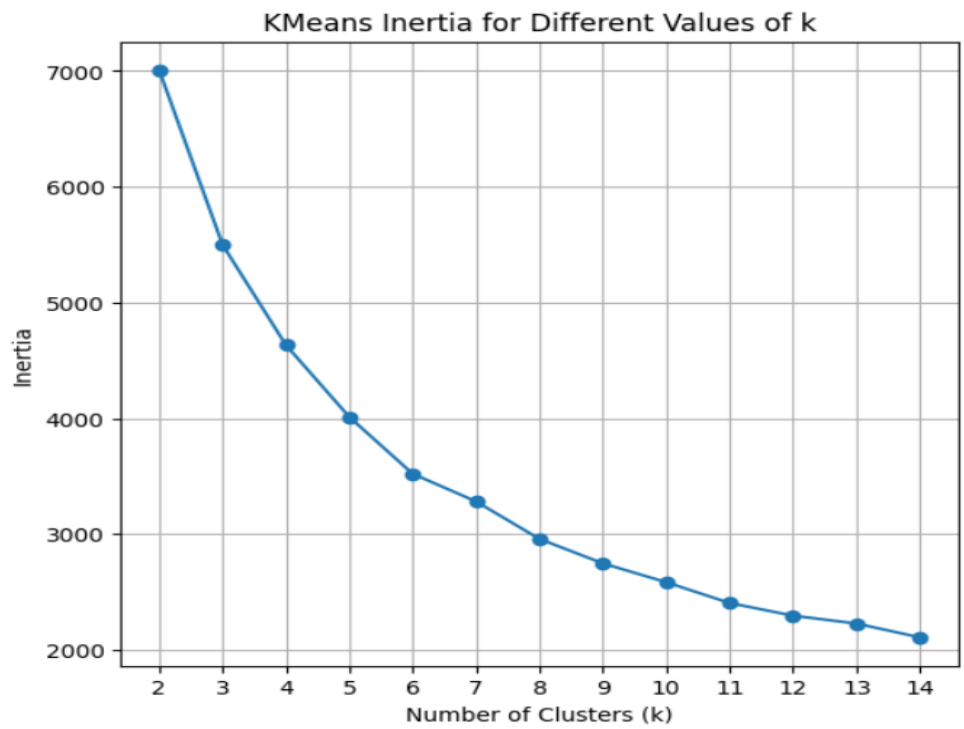

Figure 2: K Means Inertia for Different K Values

Inspection of the inertia curve revealed a clear change in slope at **K=3**. Beyond this point, the reduction in inertia became marginal, which indicates that additional clusters did not substantially improve within-cluster homogeneity. This stabilization suggests that the dominant structure in the data is captured with three clusters and further increases in K could fragment existing groups rather than uncovering meaningful patterns.

Silhouette Analysis:

The Silhouette Score provides a normalized measure of how well each observation fits within its assigned cluster compared to neighboring clusters. It is formally defined as:

$$s(i) = \frac{b(i) - a(i)}{\max\{a(i), b(i)\}}$$

where:

- a(i) is the average distance between observation i and all other points in the same cluster.
- b(i) is the minimum average distance between observation i and points in the nearest alternative cluster.

The score ranges from −1 to 1, with higher values indicating better-defined clustering.

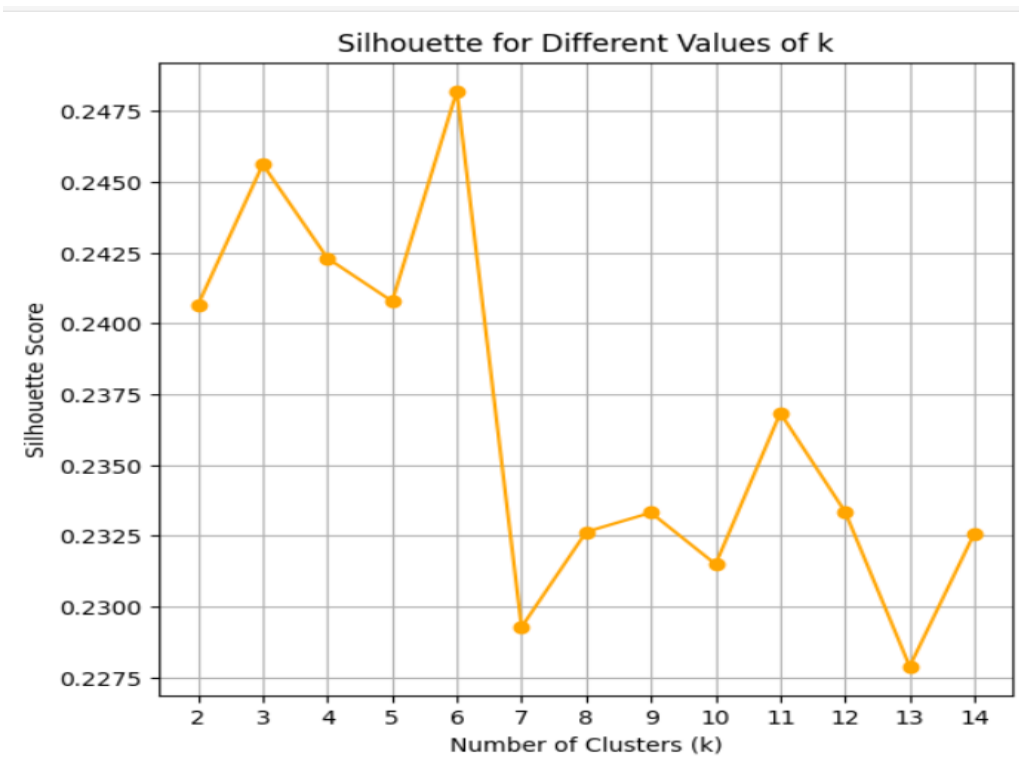

Figure 3: Silhouette for Different K Values

Silhouette analysis showed that K=6 achieved the highest overall score, while K=3 yielded the second-highest value. Importantly, the silhouette score at **K=3** (remained high) indicates well-separated clusters while avoiding excessive fragmentation of the data.

Final Selection of K:

The selection of **K=3** corresponds to the first stable point observed on the inertia curve while also exhibiting strong silhouette performance. This decision represents a balance between statistical optimality and analytical interpretability, a trade-off that is appropriate in behavioral research, where the practical meaningfulness of clusters is as important as numerical criteria.

Visual Diagnostics:

A three-dimensional scatter plot of the standardized feature space, with points colored by cluster assignment, enabled direct inspection of cluster separation and overlap. By illustrating both the relative positioning of centroids and the dispersion of observations within each cluster, the plot also reinforced conclusions drawn from inertia and silhouette analyses.

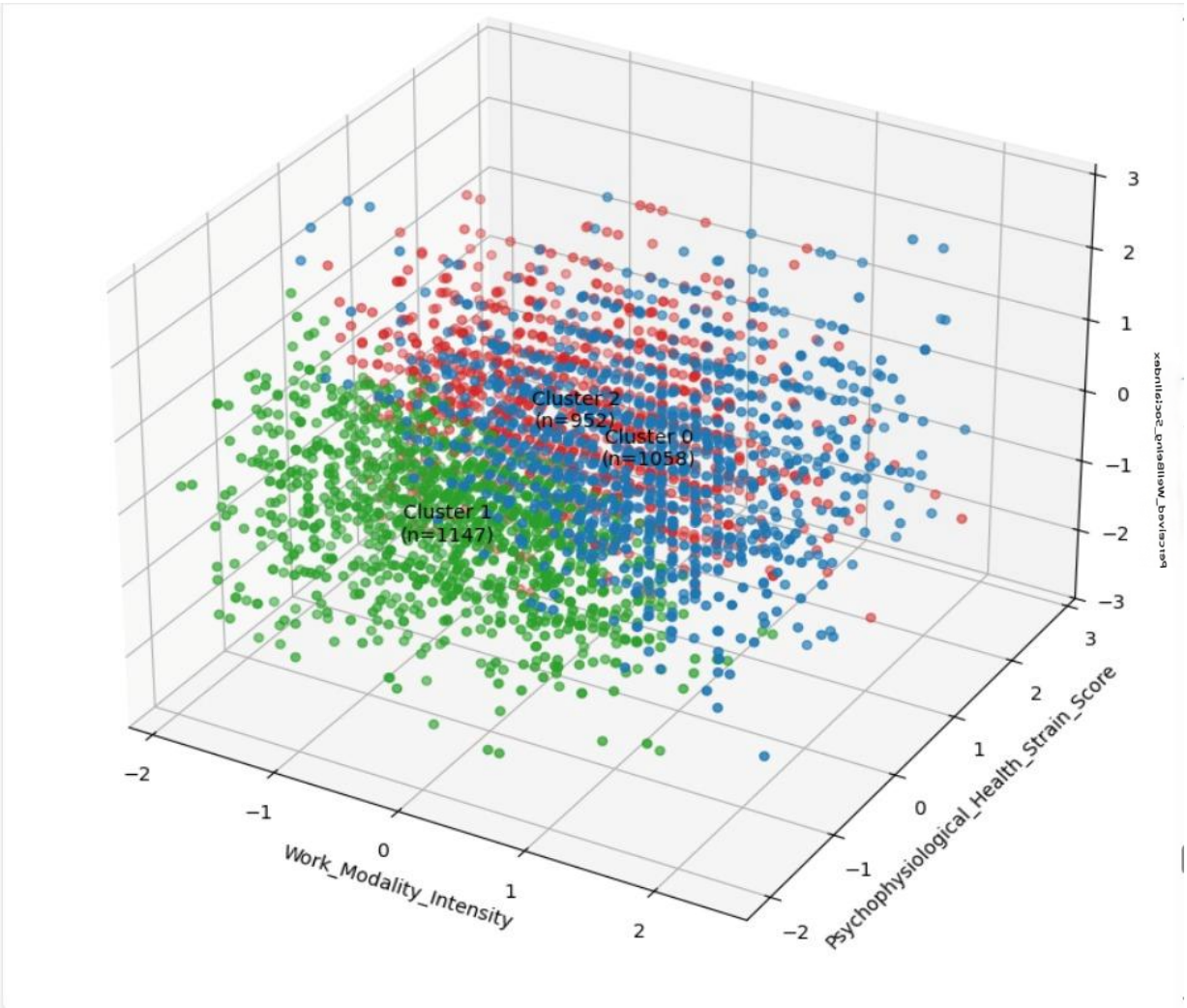

Figure 4: 3D Scatter Plot of Employee Data by Cluster

9. CLUSTER LABELING

Given the unsupervised nature of the analysis, labels were derived post hoc by examining cluster-wise distributions using violin plots that capture both central tendency and dispersion. This approach ensures that cluster names reflect dominant behavioral patterns observed in the data rather than subjective assumptions.

Cluster Size:

The final clustering solution (K=3) partitions the dataset into three relatively balanced groups:

- **Cluster 0**: 1,058 observations
- **Cluster 1**: 1,147 observations
- **Cluster 2**: 952 observations

Feature Distributions:

Cluster interpretation was informed by violin plots illustrating the distribution of:

- Work_Modality_Intensity
- Psychophysiological_Health_Strain_Score
- Perceived_WellBeing_SocialIndex

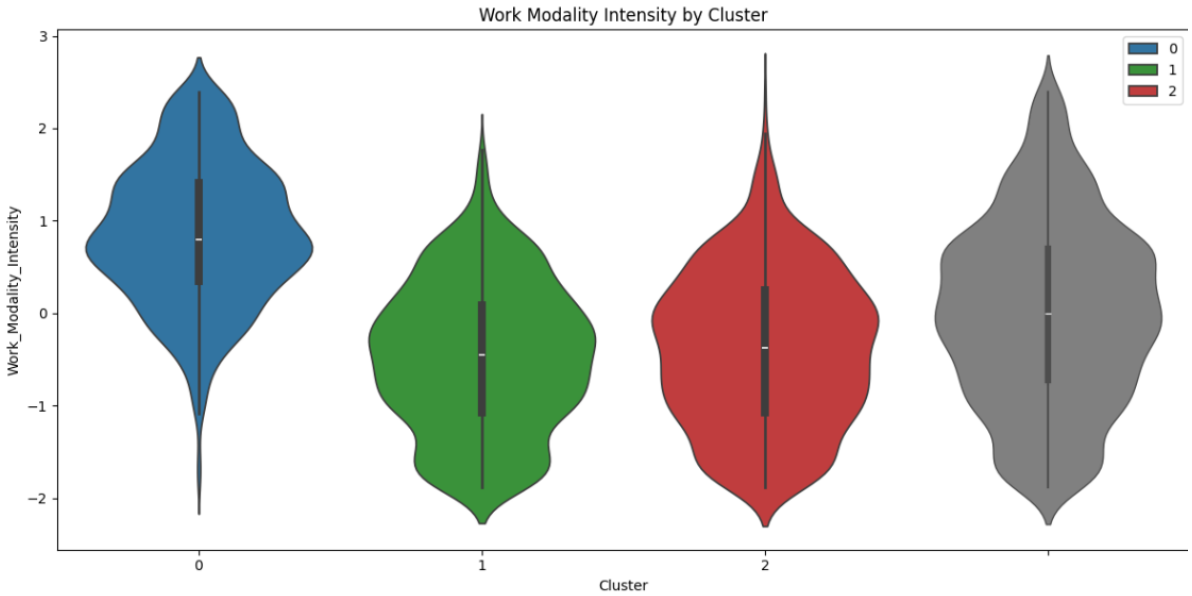

Figure 5: Work Modality Intensity by Cluster

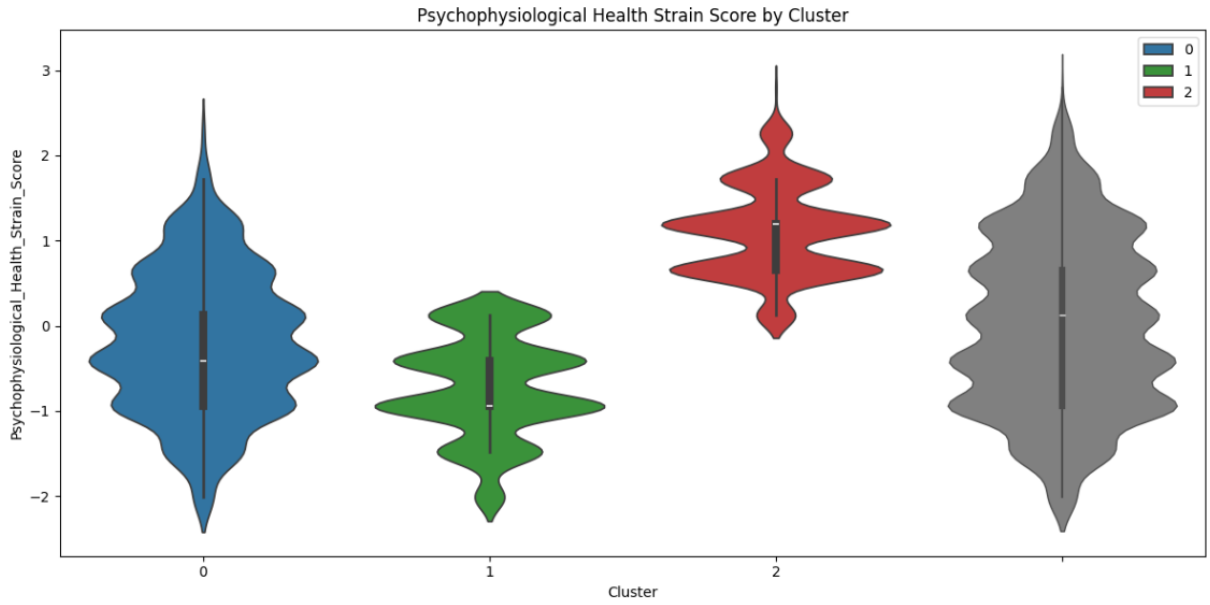

Figure 6: Psychophysiological Health Strain Score by Cluster

Figure 7: Perceived Well Being Social Index by Cluster

Cluster Profiles:

- **Cluster:** 0
- **Cluster label:** Thriving

This cluster is characterized by the **highest levels of work modality intensity and social well-being**, coupled with **low psychophysiological health strain**. The distribution of health strain scores is concentrated at lower values, which suggests resilience despite elevated work engagement.

The combination of high engagement, strong perceived well-being, and controlled health strain indicates an optimal alignment between work demands and coping resources. Individuals in this cluster appear capable of sustaining high performance without adverse health outcomes.

- **Cluster:** 1
- **Cluster label:** Detached

Cluster 1 exhibits **consistently low work modality intensity and low perceived social well-being,** alongside the **lowest and most stable health strain scores**. The distributions indicate limited engagement rather than distress-driven withdrawal.

Low strain in this context does not signal positive well-being but rather reflects minimal exposure to work-related pressure. This pattern suggests disengagement rather than balance.

- **Cluster:** 2
- **Cluster label:** Vulnerable

Cluster 2 constitutes the **highest-risk profile** in the dataset. Despite only **moderate work modality intensity,** this group displays the **highest psychophysiological health strain**, with a distribution that peaks substantially above the other clusters. Simultaneously, perceived social well-being remains low.

This configuration indicates a mismatch between work demands and coping capacity. The elevated strain in response to moderate intensity suggests limited resilience mechanisms, potentially exacerbated by insufficient social support.

Comparative Summary of Cluster Characteristics:

Cluster	Work Modality Intensity	Health Strain	Social Well-Being
Thriving	High	Low	High
Detached	Low	Very Low	Low
Vulnerable	Moderate	High	Low

Practical Implications:

The clustering analysis, grounded in an anonymized dataset encompassing both subjective measures (self-reported well-being and social connectedness) and objective indicators (work hours), provides actionable insights for multiple stakeholders. For data scientists and health researchers, the results enable the identification of risk and protective factors associated with employee well-being. For HR professionals and organizational leaders, the segmentation offers a framework to benchmark workforce health across industries and to design targeted interventions that address specific behavioral and psychosocial needs. Furthermore, these derived clusters support evidence-based policy development and future-of-work strategies that allows organizations to optimize employee engagement, resilience, and overall organizational performance. The multidimensional approach underscores the added value of clustering in revealing patterns that are not detectable through univariate analysis alone.

Limits of Interpretation:

It is critical to emphasize that cluster labels do not imply fixed individual traits or causal relationships. The analysis captures patterns within the observed feature space, and interpretations are bounded by the variables included in the clustering process. No demographic or contextual variables were used for post-hoc profiling, intentionally limiting inference to behavioral dimensions only.

10.CONCLUSION

This study applied unsupervised clustering to uncover latent employee profiles in the absence of predefined labels, focusing on productivity, mental and physical well-being, and work-life balance across onsite, hybrid, and remote work arrangements in the post-pandemic context. Using a mixed dataset of over 3,000 observations comprising numerical, ordinal, categorical, and textual variables, the analysis followed preprocessing pipeline that included exploratory data analysis, missing-value handling, appropriate encoding strategies, standardization, and feature engineering through aggregation, resulting in three composite dimensions suitable for clustering. K-Means clustering was employed, with the optimal number of clusters determined via the Elbow method and validated using silhouette evaluation, yielding three interpretable profiles: Thriving, Detached, and Vulnerable. These clusters reveal multidimensional relationships that are not observable through univariate analysis alone and provide actionable insights for data scientists, health researchers, and organizational decision-makers seeking to identify protective factors, benchmark workforce health, and design targeted interventions. Notably, cluster labels do not imply fixed individual traits or causal relationships, as interpretations are constrained to behavioral patterns within the selected feature space. Future research could enhance applicability by incorporating longitudinal data, richer contextual and behavioral variables, advanced feature extraction techniques, and alternative clustering algorithms to assess stability over time.